

ChangeVLN: A Benchmark for Vision-Language Navigation in Changed Environments

Sohee Kang and Nuri Kim*

Abstract—Vision-language navigation (VLN) benchmarks evaluate an agent against a reference trajectory in a fixed scene, implicitly assuming that this trajectory remains feasible throughout navigation. A path-blocking scene change can invalidate this assumption while leaving the instruction, start position, and goal unchanged. In this setting, reaching the goal may require a detour that necessarily deviates from the original reference trajectory, making fixed-reference evaluation insufficient for characterizing post-change navigation behavior. We introduce ChangeVLN, a benchmark for controlled path-blocking scene changes that preserves the navigation task while making the original route infeasible and retaining at least one feasible alternative path. The benchmark contains 1,317 valid episodes across 11 scenes. We further introduce a change-aware evaluation protocol that considers goal completion, path efficiency under the changed topology, compatibility with feasible post-change routes, and goal-directed progress. Experiments with a pretrained VLN policy on 500 episodes show limited post-change navigation performance and that endpoint and route-level measures provide different views of navigation behavior after environmental changes.

I. INTRODUCTION

Vision-language navigation (VLN) requires an embodied agent to follow a natural-language instruction and navigate to a target location in an unseen environment [1]. A range of VLN methods have been developed under this formulation [2]–[6]. Standard VLN benchmarks associate each instruction with a reference trajectory and evaluate navigation in a fixed scene configuration [1], [7]. Standard VLN evaluation therefore presumes that the reference trajectory remains feasible and can serve as a consistent basis for evaluating the agent trajectory.

Environmental changes can break this assumption without changing the underlying navigation objective. As illustrated in Fig. 1, the original instruction leads the agent along a valid route in the clean environment. After an obstacle is introduced, however, that route becomes blocked even though the instruction, start position, and goal remain unchanged. Reaching the same goal therefore requires the agent to adapt its route to the changed environment rather than continue along the path that was appropriate before the scene change.

This route adaptation also changes how navigation behavior should be evaluated. Once the original reference trajectory becomes infeasible, deviation from that trajectory

This work was supported by the National IT Industry Promotion Agency (NIPA) grant funded by the Korea government (MSIT) (No. RS-2026-25621586, Development and Demonstration of a Manufacturing Operations System Based on Collaborative Intelligence Physical AI).

Sohee Kang and Nuri Kim are with the Department of Electronic Engineering, Jeonbuk National University, Jeonju, Republic of Korea.
*Corresponding author: Nuri Kim (e-mail: nuri.kim@jbnu.ac.kr).

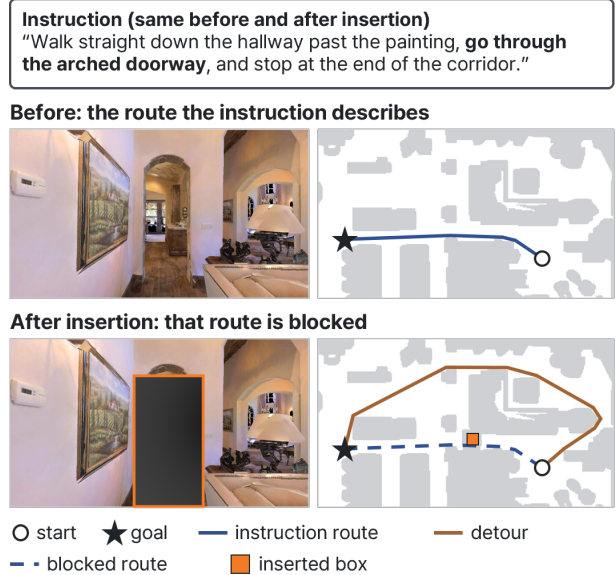


Fig. 1. A path-blocking scene change makes the instruction route infeasible while the navigation task remains the same. The instruction (top) is identical before (middle) and after (bottom) box insertion, and reaching the same goal requires a detour.

does not necessarily indicate navigation error; it may instead be required to reach the goal. A trajectory metric defined with respect to the pre-change reference can therefore penalize a successful detour. Conversely, an endpoint metric such as success indicates whether the agent reaches the goal but does not characterize how the agent responds to the changed route structure. Consequently, goal completion and agreement with the original reference capture different aspects of navigation behavior after an environmental change.

R2R-UNO [8] considers a closely related setting by introducing unexpected route obstructions and studying policy adaptation under such changes. Our work addresses a complementary question: how should navigation be evaluated when a scene change invalidates the original reference trajectory while leaving the navigation goal reachable? In this setting, evaluation should account for the route structure available after the change rather than rely exclusively on a reference trajectory that the agent can no longer follow.

We introduce ChangeVLN, a benchmark for evaluating VLN under controlled path-blocking scene changes. For each episode, ChangeVLN preserves the natural-language instruction, start position, and goal while introducing an obstacle that makes the original reference trajectory infeasible. An episode is retained only when the same goal

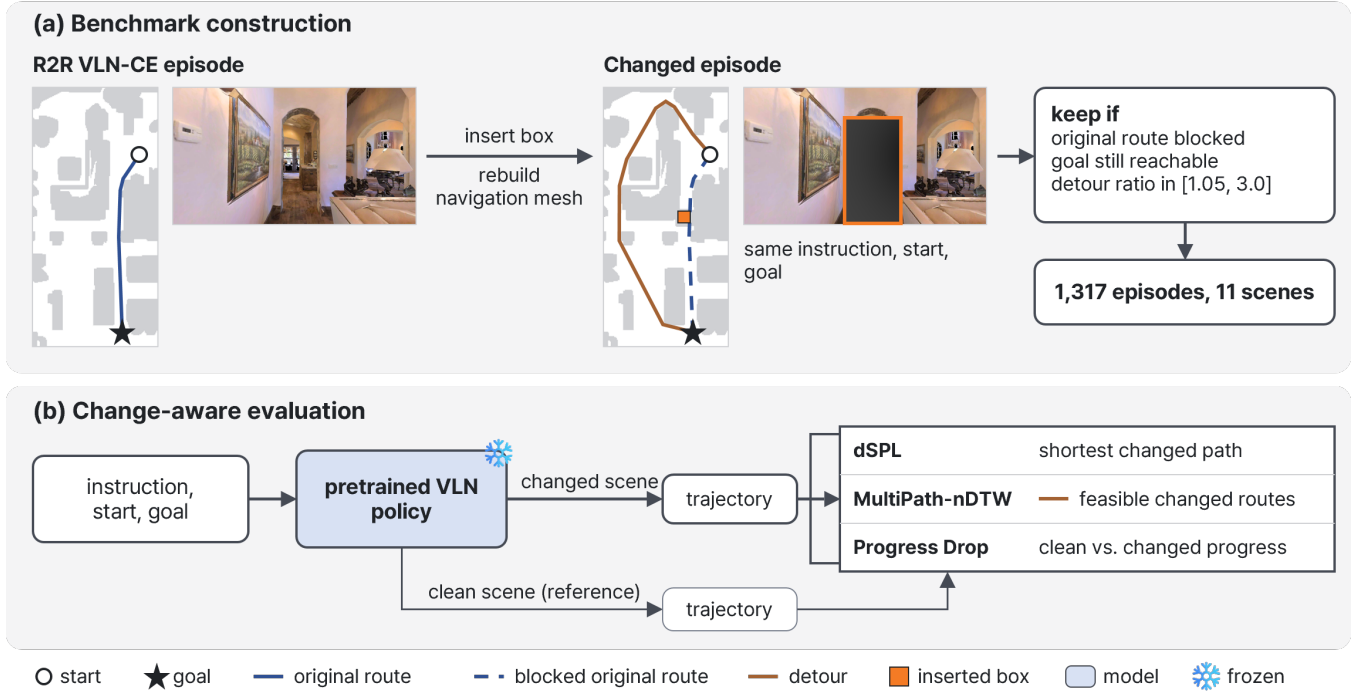


Fig. 2. ChangeVLN blocks the original route while preserving the navigation task. (a) A box is inserted along the original route, and an episode is kept only if the goal remains reachable with a detour ratio in $[1.05, 3.0]$. (b) A frozen VLN policy is run in the clean and changed scenes, and change-aware measures are computed from its trajectories.

remains reachable through at least one geometrically feasible alternative path. Applying this construction to the R2R VLN-CE val-unseen split yields 1,317 valid episodes across 11 scenes.

To evaluate navigation in these changed environments, we further introduce a change-aware evaluation protocol that complements goal-based evaluation with measures of path efficiency under the changed topology, compatibility with feasible post-change routes, and goal-directed progress. These measures distinguish whether an agent reaches the goal from how it navigates after the original route becomes unavailable. We additionally examine a lightweight obstacle-aware instruction reformulation as a controlled intervention, without modifying or retraining the navigation policy. Experiments with a pretrained VLN policy show limited post-change navigation performance and illustrate that endpoint success alone does not fully characterize post-change navigation behavior.

Our contributions are threefold:

- We introduce ChangeVLN, a benchmark for controlled path-blocking scene changes that preserve the navigation task while requiring feasible detours.
- We propose a change-aware evaluation protocol that assesses post-change navigation using goal-, trajectory-, and progress-based measures.
- We analyze a pretrained VLN policy under clean and changed environments.

II. METHOD

Fig. 2 provides an overview of ChangeVLN, which consists of benchmark construction and change-aware evaluation. First, we construct changed episodes by inserting a path-blocking obstacle while preserving the instruction, start position, and goal, and retain episodes in which the goal remains reachable through an alternative route. Second, we evaluate post-change navigation using complementary measures of goal completion, path efficiency, compatibility with feasible alternative routes, and goal-directed progress.

A. Benchmark Construction

We construct ChangeVLN from all 1,839 episodes in the R2R VLN-CE v1 val-unseen split [7]. Habitat-Sim [9] is used for obstacle insertion, navigation-mesh reconstruction, and geodesic path computation, and Habitat-Lab provides the episode and environment interfaces used for navigation.

For each episode, we insert a $0.4 \times 1.2 \times 0.4$ m box along the original reference trajectory. The navigation mesh is reconstructed after insertion so that reachability and geodesic distances reflect the modified scene geometry. Candidate obstacle placements are generated using an episode-specific fixed random seed to ensure reproducibility.

An episode is retained only if three conditions are satisfied. First, the original start-to-goal route is traversable before obstacle insertion and becomes infeasible after navigation-mesh reconstruction. Second, the original goal remains reachable in the changed environment through at least one alternative path. Third, the resulting route change must be sufficiently

large to require a meaningful detour without substantially altering the original navigation task.

Let L_{clean} and L_{chg} denote the shortest start-to-goal geodesic distances in the clean and changed environments, respectively. We define the detour ratio as $r_{\text{detour}} = L_{\text{chg}}/L_{\text{clean}}$ and retain episodes satisfying $1.05 \leq r_{\text{detour}} \leq 3.0$. The lower bound removes scene changes that produce negligible changes in the shortest path, and the upper bound excludes cases in which obstacle insertion excessively distorts the original route structure.

Applying this procedure to all 1,839 source episodes yields 1,317 valid episodes across 11 scenes, corresponding to a validity rate of 71.6%. The resulting detour ratios range from 1.050 to 2.842, with a mean of 1.199. For reproducibility, each accepted episode stores its source episode identifier, obstacle pose, clean and changed geodesic distances, detour ratio, random seed, and reconstructed navigation-mesh information.

B. Change-Aware Evaluation Protocol

A path-blocking scene change makes the original reference trajectory infeasible even though the navigation goal remains reachable. Evaluation based only on the pre-change reference may therefore penalize trajectories that successfully adapt to the modified environment. We evaluate post-change navigation along four complementary aspects: goal completion, path efficiency under the changed topology, compatibility with feasible alternative routes, and goal-directed progress.

The success rate (SR) measures whether the agent reaches the goal. We additionally use the following measures to characterize how navigation behavior changes after obstacle insertion.

Detour-Aware SPL (dSPL). SPL evaluates successful navigation relative to a shortest-path distance [7]. After obstacle insertion, the shortest feasible route can differ from that of the clean environment. We therefore compute path efficiency using the shortest geodesic distance available in the changed environment.

For episode i , let $S_i \in \{0, 1\}$ denote the success indicator, P_i the executed path length, and $L_{\text{chg},i}$ the shortest start-to-goal geodesic distance after obstacle insertion. We define

$$\text{dSPL} = \frac{1}{N} \sum_{i=1}^N S_i \frac{L_{\text{chg},i}}{\max(L_{\text{chg},i}, P_i)}. \quad (1)$$

Thus, dSPL measures efficiency relative to the route structure available to the agent after the scene change rather than to the pre-change topology.

MultiPath-nDTW. Conventional nDTW evaluates trajectory similarity with respect to a single reference trajectory [7]. Once the original route is blocked, however, several geometrically feasible paths may lead to the same goal. We therefore compare the executed trajectory with a set of reference paths generated on the changed navigation mesh.

Let

$$\mathcal{T}^* = \{\tau_1^*, \dots, \tau_K^*\} \quad (2)$$

denote the set of feasible post-change reference paths. For an executed trajectory τ , we define

$$\text{MultiPath-nDTW}(\tau) = \max_{\tau_k^* \in \mathcal{T}^*} \text{nDTW}(\tau, \tau_k^*). \quad (3)$$

For each episode, $\mathcal{T}^*$ contains one direct shortest path and up to six deterministic via-point detours generated on the changed navigation mesh. The candidate paths are resampled at a common spatial interval, and near-duplicate paths are removed before evaluation. MultiPath-nDTW therefore measures compatibility with geometrically feasible post-change routes rather than requiring agreement with the blocked pre-change trajectory.

Progress Drop. Endpoint success does not distinguish between failures that make substantial progress toward the goal and failures that remain close to the starting position. We therefore measure the reduction in geodesic distance to the goal.

For episode i , let $d_{0,i}$ denote the start-to-goal geodesic distance and $d_{T,i}$ the geodesic distance from the agent's final position to the goal, both computed on the navigation mesh corresponding to the evaluated condition. We define goal progress γ_i as

$$\gamma_i = \max\left(0, \frac{d_{0,i} - d_{T,i}}{d_{0,i}}\right). \quad (4)$$

The progress drop caused by the scene change is defined as

$$\Delta_i = \gamma_i^{\text{clean}} - \gamma_i^{\text{changed}}. \quad (5)$$

We report the mean progress drop across episodes. Positive values indicate reduced final goal-directed progress after obstacle insertion.

All geodesic distances are computed using Habitat-Sim's navigation-mesh pathfinder on the navigation mesh corresponding to each condition. MultiPath-nDTW is intentionally based on geometric feasibility and does not guarantee that every alternative reference satisfies all spatial or semantic constraints expressed in the instruction. We examine this distinction through the semantic consistency analysis in Sec. III-D.

III. EXPERIMENTS

A. Experimental Setup

We evaluate the pretrained NaVILA-Llama3-8B-8f policy [10] without additional training. Each selected episode is evaluated in both the clean and changed environments using the same instruction, start position, and goal. An episode terminates when the policy issues a STOP action or reaches the maximum horizon of 1,000 steps.

From the 1,317 valid ChangeVLN episodes, we evaluate 500 episodes. The evaluation set contains 91 episodes retained from preliminary experiments and 409 additional episodes sampled from the remaining benchmark episodes using a fixed random seed of 42. The additional episodes are sampled according to scene and detour-ratio interval to provide broad coverage of the benchmark distribution. We

TABLE I
NAVIGATION PERFORMANCE IN CLEAN
AND PATH-BLOCKED ENVIRONMENTS.

Env.	Method	Standard		Change-aware		
		SR $\uparrow$	nDTW $\uparrow$	dSPL $\uparrow$	MultiPath-nDTW $\uparrow$	Progress Drop $\downarrow$
Clean	NaVILA [10]	54.0	–	49.0	–	–
Path-blocked	NaVILA [10]	19.80	43.55	17.29	51.36	26.54
	NaVILA [10] + OAP	20.80	41.62	16.95	50.64	22.05

use six detour-ratio intervals: $[1.0, 1.1)$, $[1.1, 1.2)$, $[1.2, 1.4)$, $[1.4, 1.6)$, $[1.6, 2.0)$, and $[2.0, 10.0]$.

We additionally evaluate an obstacle-aware prefix (OAP) as a lightweight instruction-level intervention. OAP prepends the fixed statement “There may be an obstacle blocking your path. If blocked, carefully turn and navigate around it, then continue:” to the original navigation instruction. The same pretrained policy is used without additional training or policy modification, and all other evaluation settings remain unchanged.

B. Main Evaluation Results

Table I compares NaVILA in the clean environments with NaVILA under path-blocking scene changes, using the evaluation protocol introduced in Sec. II-B. The clean-environment row reports the R2R VLN-CE val-unseen results published for NaVILA [10], and the other rows are computed on our 500 episodes. In the clean environments, dSPL is identical to SPL because the shortest path is unchanged, and MultiPath-nDTW and Progress Drop are not defined; the NaVILA paper does not report nDTW on R2R VLN-CE. Shaded cells mark the better result between the two path-blocked rows.

NaVILA reports a success rate of 54.0% and an SPL of 49.0 in the clean environments. Under path-blocking scene changes, its success rate decreases to 19.80% and its dSPL to 17.29, and Progress Drop reaches 26.54 relative to the corresponding clean-condition executions. Because the published results cover the full split, we also compare our own clean and changed runs on the same episodes: on the 409 newly sampled episodes, for which clean-environment trajectories are stored, the success rate decreases from 55.50% to 22.74%. These results indicate limited goal-directed navigation performance after the original route becomes infeasible, even though the instruction and goal remain unchanged.

NaVILA obtains an nDTW of 43.55 against the original reference trajectory and a MultiPath-nDTW of 51.36 against geometrically feasible post-change references. Because the two metrics use different reference sets, their absolute values should not be interpreted as a direct metric-to-metric comparison. Instead, the difference illustrates how trajectory-level assessment depends on whether evaluation remains tied to the blocked pre-change reference or accounts for routes that remain feasible after the scene change.

C. Obstacle-Aware Instruction Analysis

OAP increases the observed success rate from 19.80% to 20.80%, corresponding to 99 and 104 successful episodes out of 500, respectively. This increase is not accompanied by consistent improvement in the other post-change measures. dSPL decreases from 17.29 to 16.95, nDTW from 43.55 to 41.62, and MultiPath-nDTW from 51.36 to 50.64.

At the episode level, 44 failures become successes whereas 39 successes become failures; a two-sided exact McNemar test shows that this difference is not statistically significant ($p = 0.661$). OAP also reduces Progress Drop from 26.54 to 22.05. Because Progress Drop is measured relative to the clean-condition performance of each instruction condition, this reduction does not by itself indicate better post-change navigation: OAP lowers the mean clean-condition goal progress from 55.36 to 50.78, whereas the mean changed-condition goal progress remains at 28.82 and 28.74. OAP also does not improve path efficiency or trajectory compatibility. We therefore treat OAP as an exploratory intervention for examining sensitivity to obstacle-related linguistic context rather than as evidence of a reliable performance improvement.

D. Semantic Consistency of Alternative References

MultiPath-nDTW evaluates trajectories against geometrically feasible post-change references, but geometric feasibility does not necessarily imply consistency with all spatial constraints expressed in the instruction. We conduct a preliminary analysis on 10 episodes to examine this distinction.

For each episode, room-level mentions in the instruction are matched to Habitat semantic regions, and an alternative reference is retained only when it traverses the required regions in the specified order. Geometrically feasible references are available for all 10 episodes, whereas only five episodes retain at least one reference after applying these region constraints.

Although the sample is too small to support a general quantitative conclusion, the result indicates that geometric feasibility and instruction consistency need not coincide. MultiPath-nDTW should therefore be interpreted as a measure of compatibility with feasible geometric alternatives rather than as a complete measure of instruction fidelity.

IV. CONCLUSION

We introduced ChangeVLN, a benchmark and evaluation protocol for vision-language navigation under controlled path-blocking scene changes that preserve the navigation task while making the original reference trajectory infeasible. Experiments with a pretrained VLN policy show limited post-change navigation performance and demonstrate that goal-, trajectory-, and progress-based measures capture complementary aspects of navigation behavior. Our study is limited to a single policy, one fixed box obstacle per scene change, and a single dataset split, and the geometrically feasible alternative paths may not preserve all spatial constraints in the instruction, motivating future evaluation protocols that incorporate semantic consistency.

REFERENCES

- [1] P. Anderson, Q. Wu, D. Teney, J. Bruce, M. Johnson, N. Sünderhauf, I. Reid, S. Gould, and A. van den Hengel, “Vision-and-Language Navigation: Interpreting Visually-Grounded Navigation Instructions in Real Environments,” in *Proc. of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, Salt Lake City, UT, USA, Jun. 2018.
- [2] D. Fried, R. Hu, V. Cirik, A. Rohrbach, J. Andreas, L.-P. Morency, T. Berg-Kirkpatrick, K. Saenko, D. Klein, and T. Darrell, “Speaker-Follower Models for Vision-and-Language Navigation,” in *Proc. of Advances in Neural Information Processing Systems (NeurIPS)*, Montréal, Canada, Dec. 2018.
- [3] W. Hao, C. Li, X. Li, L. Carin, and J. Gao, “Towards Learning a Generic Agent for Vision-and-Language Navigation via Pre-Training,” in *Proc. of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Seattle, WA, USA, Jun. 2020.
- [4] Y. Hong, Q. Wu, Y. Qi, C. Rodriguez-Opazo, and S. Gould, “VLN BERT: A Recurrent Vision-and-Language BERT for Navigation,” in *Proc. of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Virtual Conference, Jun. 2021.
- [5] S. Chen, P.-L. Guhur, C. Schmid, and I. Laptev, “History Aware Multimodal Transformer for Vision-and-Language Navigation,” in *Proc. of Advances in Neural Information Processing Systems (NeurIPS)*, Virtual Conference, Dec. 2021.
- [6] S. Chen, P.-L. Guhur, M. Tapaswi, C. Schmid, and I. Laptev, “Think Global, Act Local: Dual-Scale Graph Transformer for Vision-and-Language Navigation,” in *Proc. of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, New Orleans, LA, USA, Jun. 2022.
- [7] J. Krantz, E. Wijmans, A. Majumdar, D. Batra, and S. Lee, “Beyond the Nav-Graph: Vision-and-Language Navigation in Continuous Environments,” in *Proc. of the European Conference on Computer Vision (ECCV)*, Glasgow, UK, Aug. 2020.
- [8] H. Hong, S. Wang, Z. Huang, Q. Wu, and J. Liu, “Navigating Beyond Instructions: Vision-and-Language Navigation in Obstructed Environments,” in *Proc. of the 32nd ACM International Conference on Multimedia (ACM MM)*, Melbourne, Australia, Oct. 2024.
- [9] M. Savva, A. Kadian, O. Maksymets, Y. Zhao, E. Wijmans, B. Jain, J. Straub, J. Liu, V. Koltun, J. Malik, D. Parikh, and D. Batra, “Habitat: A Platform for Embodied AI Research,” in *Proc. of the IEEE/CVF International Conference on Computer Vision (ICCV)*, Seoul, Republic of Korea, Oct. 2019.
- [10] A.-C. Cheng, Y. Ji, Z. Yang, Z. Gongye, X. Zou, J. Kautz, E. Bıyık, H. Yin, S. Liu, and X. Wang, “NaVILA: Legged Robot Vision-Language-Action Model for Navigation,” in *Proc. of Robotics: Science and Systems (RSS)*, Los Angeles, CA, USA, Jun. 2025.